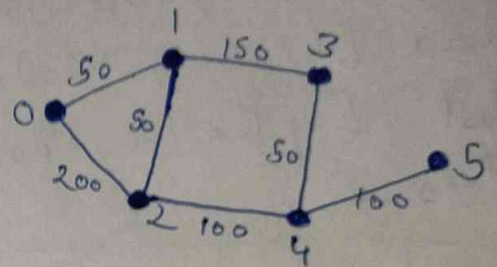


Dry run (Example)

$n = 6$

flights = {
 {0, 1, 50},
 {0, 2, 200},
 {1, 2, 50},
 {1, 3, 150},
 {2, 4, 100},
 {3, 4, 50},
 {4, 5, 100}}
}

src = 0, dst = 5, K = 2



Dry Run

• Initial state

dist = [0, ∞, ∞, ∞, ∞, ∞]

queue = [(0, 0)]

steps = 0

• Level 0 (0 steps)

From node 0:

• 0 → 1 = 50

• 0 → 2 = 200

temp = [0, 50, 200, ∞, ∞, ∞]

queue = [(1, 50), (2, 200)]

dist = [0, 50, 200, ∞, ∞, ∞]

steps = 1

Level 1 (1 stop)

From node 1

$1 \rightarrow 2 = 100$ (better than 200)

$1 \rightarrow 3 = 200$

From node 2

$2 \rightarrow 4 = 300$

temp = [0, 50, 100, 200, 300, ∞]

queue = [(2, 100), (3, 200), (4, 300)]

dist = [0, 50, 100, 200, 300, ∞]

stops = 2

Level 2 (2 stops)

From node 2:

$2 \rightarrow 4 = 200$

From node 3:

$3 \rightarrow 4 = 250$ (not better than 200)

From node 4:

$4 \rightarrow 5 = 400$

temp = [0, 50, 100, 200, 200, 400]

queue = [(4, 200), (5, 400)]

dist = [0, 50, 100, 200, 200, 400]

stops = 3 (stop, $K=2$)

dist[5] = 400

But check the path

0-1-2-4-5

This use 3 stops which is more than $K=2$

Valid path $K=2$

0-2-4-5

Cost = $200 + 100 + 100 = 400$

stops = 2 ✓

Final
answer 400